

Ayush Raj

Third Year Undergraduate
Department of Aerospace Engineering

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🌐 Ayush Raj | 📄 Ayushraj6942

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	6.6/10
2023	CBSE(XII)	Indian Convent School,Kota	94.2%
2021	CBSE(X)	Army Public School,Sri Ganganagar	96.4%

Scholastic Achievements

- Secured All India Rank **3946** in **JEE Advanced 2023** among 2.5 lakh shortlisted candidates.
- Secured All India Rank **4175** in **JEE Mains 2023** among 11 lakh candidates.

Research Experience

SURGE’25 IIT KANPUR | B2B Brand Equity | Prof.RN Sengupta (May’25-July’25)

Objective	• Identify determinants of B2B brand equity using literature, company data, and test models via ML .
Approach	• Reviewed literature on B2B Brand Equity to map constructs like awareness, loyalty, trust, and reputation. • Applied SEM and multiple logistic regression on 5 firms; analyzed impact of digital marketing using MROI . •Analysed Semantic Brand Score to measure brand importance and strength using textual data and Monte Carlo Simulation for Brand Equity Risk Optimization
Impact	• Proposed methods to quantify brand equity using ML models and highlighted digital marketing’s role via MROI .

Key Projects

White Paper For Assistive Devices | IITK Consulting Group | Department Of SnT 🌀 (April’25-June’25)

- Worked under **Dr. Anita Aggarwal (DST, NAMEDD)** on a national white paper for assistive devices (ADs), targeting policy investment.
- Co-authored recommendations using global best practices comparative insights from **USA, Japan, Germany, Denmark**.
- Built global leaderboard, mapped **India’s top 10 medtech states**; analyzed **7+ govt schemes, 20+ VC firms/accelera-**

Course Project-DMS624: Computer-Aided Decision Systems | Prof. Deepu Philip 🌀 (Aug’25-Nov’25)

- OrbitWear: End-to-End Merchandise Decision System**
 - Designed **end-to-end merchandise solution** (design, pricing, delivery) for IITK clubs, resolving vendor coordination issues.
 - Managed **pricing, advance payments, vendor payouts, and profit tracking** under a low-capital model to minimize financial risk.
- Solving Farmers’ Loan Access: Digital Kisan ID–Based Credit Framework**
 - Analyzed credit access barriers using **NABARD, RBI, & PM-KISAN reports** to identify systemic bottlenecks.
 - Proposed **Digital Kisan ID** integrating Aadhaar & land records to **auto-fill 80% of loan forms** for instant application.

TaxSmart | Capstone Project | Product Club IITK 🌀 (Dec’25)

- Conceptualized a **proactive tax planning product** addressing fragmented data, low tax awareness, and last-minute filing behavior.
- Designed an **MVP** including a Tax Planner Quiz, personalized tax plan, document vault, year-round tracking, and a unified Tax Health Score.
- Conducted competitive analysis of **Groww, ClearTax, and ET Money**; identified personalization gaps and proposed a product-led {GTM with content-led onboarding, reminder-based retention, and **ELSS/NPS/insurance** cross-sell.

Positions of Responsibility

Business Head | IITK Rocketry & Space Exploration Team (Mar’24–Present)

- Contributed to **avionics, structures, propulsion, recovery**; launched rocket to **600 ft** apogee in **July 2024**.
- Managed finances; secured **2 sponsorships**, handled **5 lakh** budget, and oversaw bill clearance.
- Led **30+ juniors**, handled outreach, sponsorship, and participating as Recovery member in **INSPACE Competition**.

Marketing Secretary | Udghosh’24 (Aug’24-Oct’24)

- Led lead generation and **database management** for **100+ companies** to support sponsorship and marketing outreach.
- Developed strong **communication skills** including cold calling, email writing, and persuasive brand negotiation.

Hospitality Secretary | Inter IIT Sports Meet’24 (Nov’24-Dec’24)

- Managed hospitality for over **5,000 players** across IITs, ensuring smooth accommodation, meals, and logistics.
- Utilized **Microsoft Excel** for rooming, food tracking, and scheduling to improve efficiency and reduce errors.

Extra-Curricular Activities

- Completed **NCC ‘C’ Certificate** program, showcasing leadership, discipline, commitment, and physical endurance.
- Represented Team RaSET at **IITK Samanvay: Industry Engagement Event 2025**, showcasing rocketry work to industry and academia.
- Contributed to **Antarixx’25 – National Space Day Celebration**, conducting live demos and science outreach on space.
- Participated in and cleared **PDR and CDR** rounds of **INSPACE Model Rocketry Competition** with IITK RaSET team.